

TraceLens

Explainable code-change impact intelligence

Theme

Developer Tooling:
Understanding Code Changes
and Debugging

Core idea

Show the impact of every code
change.

Focus

Code review, impact analysis &
test coverage

Problem Statement

A small diff can hide a large risk.

A small diff

A token parser edit looks tiny, but can touch login, sessions, API clients and tests.

The friction

Reviewers trace imports by hand
Test coverage is guessed
Intent is checked manually

The central question

How can a reviewer see the real impact of a change before merging?

Small diff



Manual tracing



Blind spots



Missed tests



Regression



Rework

Existing Solutions & The Gap

PR review tools

Changed lines and comments

Gap: no codebase-wide impact view.

Linters & analyzers

Style rules and bug patterns

Gap: do not explain the reach of this change.

CI test suites

Run the tests that exist

Gap: cannot reveal tests that are missing.

AI code reviewers

Summaries and suggestions

Gap: claims may lack traceable evidence.

Identified gap

Most tools show what changed. TraceLens shows what may be affected, with evidence a developer can verify.

Proposed Solution

Give every pull request an evidence-linked impact report.

The shift

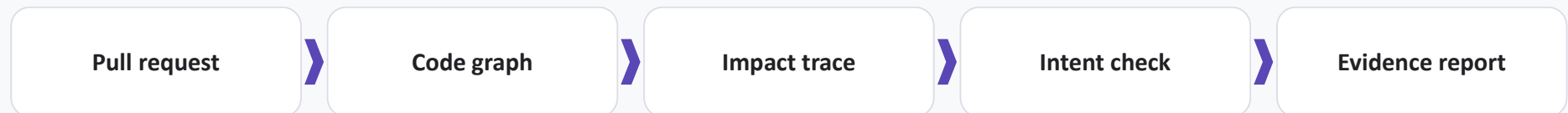
Instead of:
“I reviewed the changed lines.”

Think:
“I know which code paths and tests this change touches.”

The principle

TraceLens supplies the evidence.

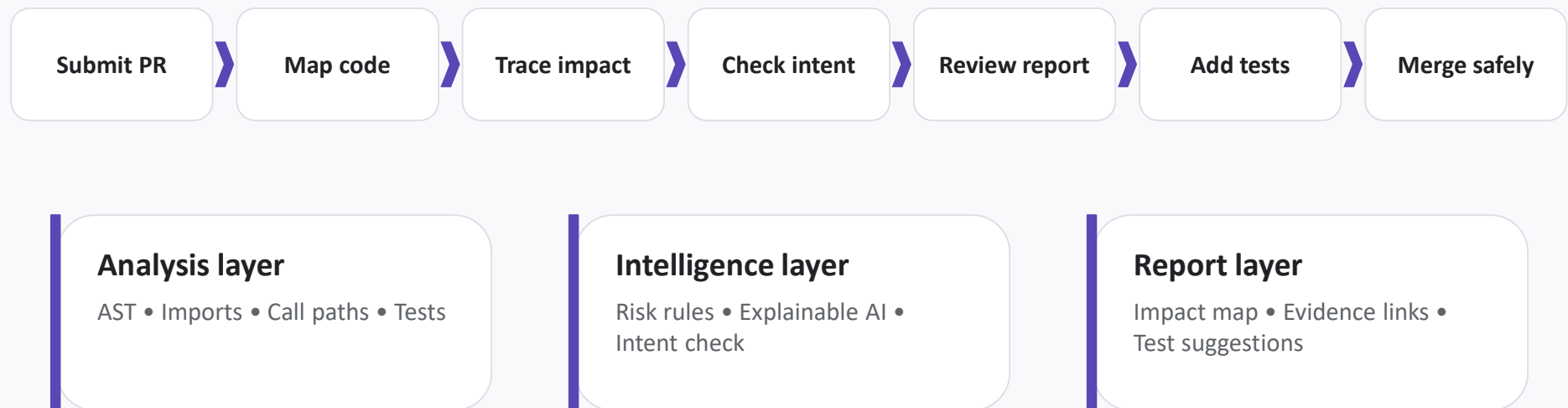
The developer still decides what to review, test and merge.



Proposed Solution

The Impact Workflow

TraceLens runs at pull-request time and gives the developer a focused review checklist.



Proposed Solution

Key Features

Diff Analysis

Reads changed files and lines from a pull request.

Code Graph

Maps symbols, imports, call paths and tests.

Impact Tracing

Finds directly and indirectly connected modules.

Risk Rules

Flags public API, auth, schema and test signals.

Test Suggestions

Points to related tests and missing edge cases.

Intent Check

Compares the change with the stated task.

Evidence Links

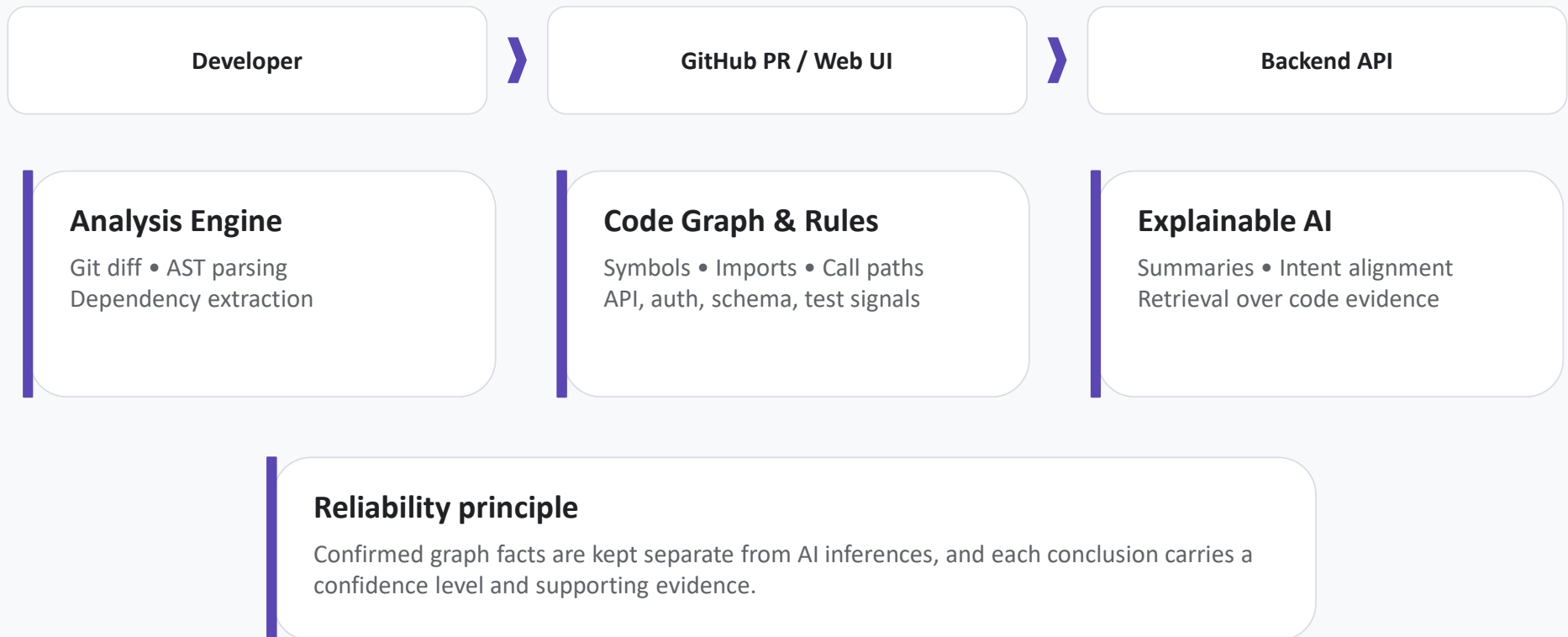
Every finding links to code, a dependency or a test.

Confidence Levels

Separates graph facts from AI inferences.

Technical Approach

Deterministic code analysis feeds an AI layer that explains only the available evidence.



Technology Stack

Technology follows the requirements of the system.

Frontend

React / Next.js — PR report and interactive impact map

Backend

Python FastAPI or Node.js — Run analysis and generate reports

Code intelligence

Git + Tree-sitter — Parse diffs, extract symbols, imports, calls and tests

Dependency graph

NetworkX or Neo4j — Store and traverse the code relationship graph

AI layer

LLM with retrieval over code evidence — Explain intent with traceability

Integration

GitHub REST / GraphQL API — Fetch PR metadata, commits and repo context

MVP scope

JavaScript / TypeScript repositories first — Diff → graph → related tests → report

Feasibility, Limitations & Impact

Feasibility

- Proven parsers (Tree-sitter, AST)
- Graph traversal with NetworkX / Neo4j
- GitHub API supplies PR data
- MVP scoped to JavaScript / TypeScript

Limitations

- Static analysis can miss dynamic calls
- AI inferences can be wrong, so each carries a confidence level
- Reports guide review; they do not replace it

Potential impact

- Student teams learning collaboration
- New open-source contributors
- Developers in unfamiliar codebases
- Reviewers short on time

Core value

Change the review loop: from “read the changed lines” to clear evidence of what is affected and what to test before merge.

The Core Idea

A different question for code review.

Traditional review

How can we review diffs faster?

Diff → Review → Merge

TraceLens approach

What if every change came with a verified map of what it touches?

Diff → Trace → Evidence → Tests → Merge

Example: PR “Update authentication token format”

auth/tokenParser.ts



loginController.ts



sessionMiddleware.ts



Missing expired-token
test



Flagged HIGH IMPACT

Thank You

Questions & Discussion

Team TraceLens

Prachi Singh (Team Leader) • Sundaram Gupta

Judge discussion

Why this problem? • Why this approach? • What are the trade-offs?

TraceLens | See beyond the diff